SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C04 - ASSESSMENT TOOL 1 – Scenario Based

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 – Scenario Based**	**Assessment:**	**Assessment Tool 1– Scenario Based**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	**Analyse and apply point estimation, frequentist estimation, and confidence interval techniques to estimate population parameters and interpret statistical results effectively. (BL4)**		
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Section / Batch:	AI&DS	**Date:**	17-08-2026

Code of Conduct

I Yedururi Anjani Priya (192324176) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Anjani Priya

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. A food delivery company wants to know the average delivery time in Chennai. It randomly selects 100 deliveries and obtains an average delivery time of 32 minutes. What population parameter is being estimated, and what is the point estimate?

Average Delivery Time – Point Estimate

A food delivery company randomly selects 100 deliveries and obtains an average delivery time of 32 minutes.
Population Parameter
The population parameter being estimated is:
Point Estimate
The sample mean is used as the point estimate of the population mean.
Answer:

- Population Parameter: Population mean delivery time
- Point Estimate: 32 minutes

2. An e-commerce website wants to estimate the percentage of customers who purchase a product after viewing its advertisement. From 800 randomly selected visitors, 176 made a purchase. Determine the point estimate of the population proportion.

E-Commerce Purchase Proportion – Point Estimate

Given:

- Total visitors selected:
- Visitors who made a purchase:

The point estimate of the population proportion is:
As a percentage:
Answer:
The estimated percentage of all visitors who purchase the product after viewing the advertisement is 22%.

3. A streaming platform selects 500 subscribers to estimate the percentage of users who watch movies every day. If 320 users report watching movies daily, calculate the point estimate and explain its interpretation.

Given:

- Sample size:
- Sample mean expenditure:

The population mean is unknown. In the frequentist approach, the sample mean is used as an estimate of the population mean.

Therefore:

Answer: The bank uses the sample mean of ₹24,500 as the frequentist point estimate of the true average monthly expenditure of all its credit-card customers. If many random samples were taken, their sample means would vary, but the sample mean is an unbiased estimator of the population mean.

4. A streaming platform selects 500 subscribers to estimate the percentage of users who watch movies every day. If 320 users report watching movies daily, calculate the point estimate and explain its interpretation.
Daily Movie Watchers – Point Estimate

Given:

- Total subscribers selected:
- Users watching movies daily:

The point estimate of the population proportion is:

As a percentage:

Answer:

Interpretation: Based on the sample, it is estimated that approximately 64% of all subscribers watch movies every day.

5. A manufacturing company randomly selects 64 smartphones and finds that their average battery life is 18.5 hours with a standard deviation of 2 hours. Construct a 95% confidence interval for the average battery life.

95% Confidence Interval for Average Battery Life

Given:

- Sample size:

- Sample mean: hours
- Sample standard deviation: hours

For a 95% confidence interval with a large sample, use:

Step 1: Calculate Standard Error

Step 2: Calculate Margin of Error

Step 3: Calculate Confidence Interval

Lower limit:18.01

Upper limit: 18.99

Final Answer

Interpretation: We are 95% confident that the true average battery life of the population of smartphones lies between 18.01 hours and 18.99 hours.